Pitch Trim System (S.O.O. 6221) – General Data

1. General

The autopilot Pitch Trim System commands elevator trim to offload loads on the elevator servo.The pitch trim servo commands are either derived from within the AIOP of the MAU using Automatic Pitch Trim commands, or manual electric pitch trim commands derived from the Pilot and Co-pilot flight control yoke pitch trim switch and relays.

The Pitch Trim System includes the following main elements:

- Automatic Pitch Trim commands (AIOP of the MAU derived commands)
- Flight Controls Split Trim Switch (Manual Electric Pitch Trim)
- PITCH TRIM Switch installed on the pilot switch panel
- KS 272A Pitch Trim Servo (includes KM 275 capstan)

2. Description

A. Flight Controls Split Trim Switch

Trim switch marked TRIM DN and TRIM UP manually controls the pitch trim servo. The trim switch has two levers (refer to Figure 1, Detail A), trim engage and trim direction. Both levers must be engaged to actuate the trim servo. Activation of the trim engage switch while auopilot is engaged will disengage autopilot and automatic pitch trim.

B. Pitch Trim Switch

A two position pitch trim lever lock master switch labelled PITCH TRIM is located on the pilot's switch panel (refer to Figure 1, Detail B). The switch has two positions DISABLE (up) and NORMAL (down). The switch must be in the NORMAL position for the autopilot to function or for manual electronic control of the elevator trim servo. Selecting DISABLE will disconnect autopilot as well as disabling the manual electric pitch trim.

C. Pitch Trim Servo (KS 272A)

The pitch trim servo contains a DC motor and drive engagement solenoid. The pitch trim servo is mounted on and drives a capstan which actuates the elevator trim tab through servo bridle cables clamped to the trim tab control cables. Torque limiting is controlled by a slip clutch within the servo capstan.

28 Volt DC power is protected by a 5 Amp PITCH TRIM (N1) (Left Bus) circuit breaker located on the avionics circuit breaker panel (refer to 24-60-00, DC Electrical Load Distribution – General Data).

3. Adjustment/Test

A. Manual Electric Pitch Trim Operational Test

1) Ensure that PITCH TRIM (N1) (Left Bus), circuit breaker on the avionics circuit breaker panel is closed.

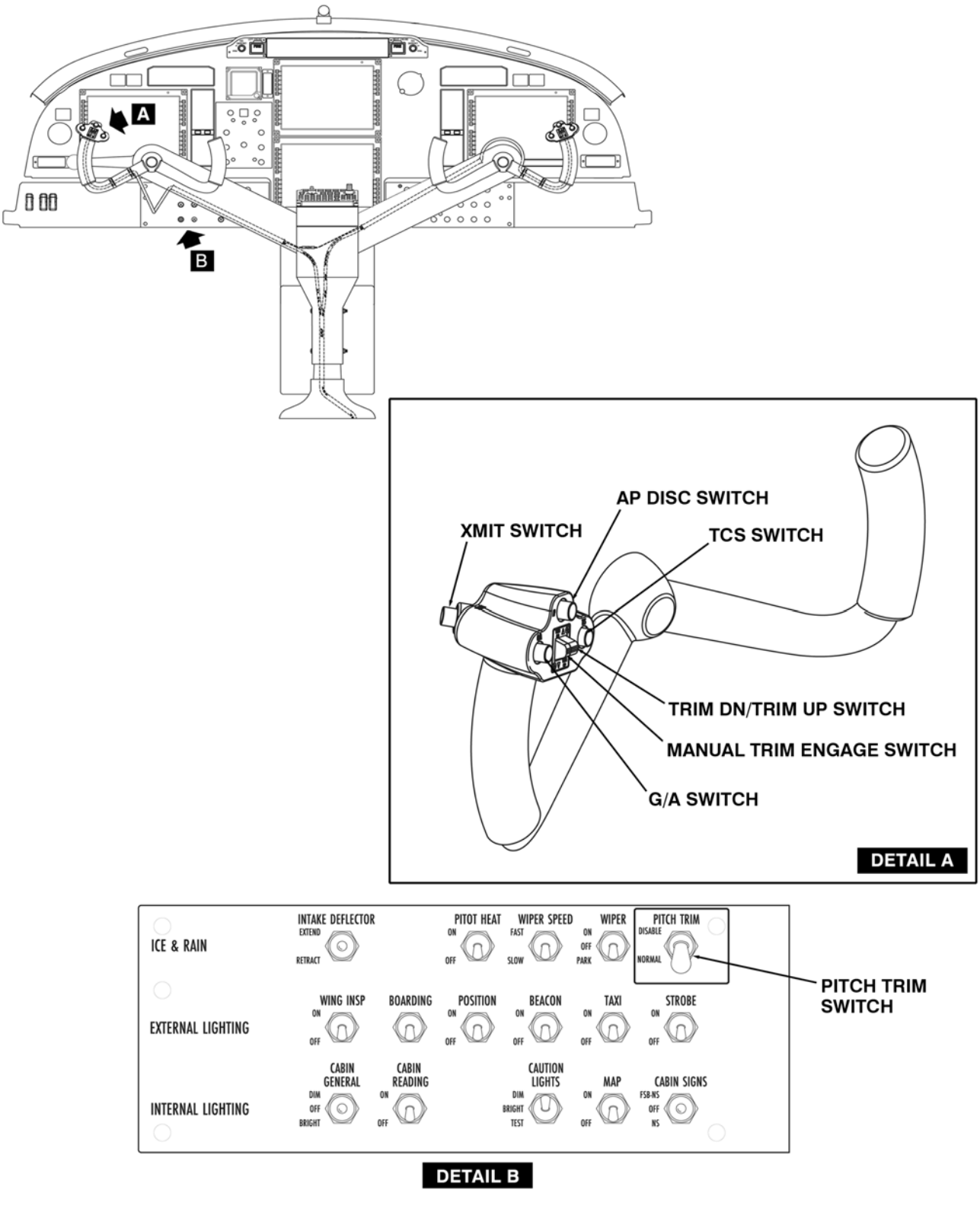
2) Ensure the PITCH TRIM switch is in the NORMAL position.

3) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4) Wait several minutes for the aircraft electrical and avionics systems to initialize. During initialization a Power-on Built In Test (PBIT) will be completed on the Pitch Trim Servo System, confirm the AP 1 Fail CAS message extinguishes after the avionics system initializes.

5) Select the Pilot's control wheel switch to the TRIM UP position, confirm the trim wheel travels TRIM UP.

6) Select the Co-pilot's control wheel switch to the TRIM UP position, confirm the trim wheel travels TRIM UP.



Pitch Trim System Controls
Figure 1

7) Select the Pilot's control wheel switch to the TRIM DN position, confirm the trim wheel travels TRIM DN.

8) Select the Co-pilot's control wheel switch to the TRIM DN position, confirm the trim wheel travels TRIM DN.

9) Select and hold a control wheel trim switch to either the TRIM UP or TRIM DN position and press and hold the AP DISC simultaneously, confirm the control wheel stops. Release the AP DISC, confirm the trim wheel travel resumes. Release the trim switch.

10) Select and hold the trim switch to either the TRIM UP or TRIM DN position and select the PITCH TRIM switch to the DISABLE position, confirm the control wheel stops. Release the trim switch. Confirm the AP 1 Fail CAS message illuminates.

11) Select the PITCH TRIM switch to the NORMAL position, confirm the AP 1 Fail CAS message extinguishes.

12) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Bench Test KM 275 Capstan

CAUTION **ANY DISASSEMBLY/ASSEMBLY OF THE SERVO MUST BE DONE AT A STATIC-SAFE WORKSTATION. THE TECHNICIAN AND BENCH SHOULD BE GROUNDED.**

1) Remove the pitch trim servo from aircraft, refer to Para 4.A.1).

2) Remove the capstan from the aircraft, refer to Para 4.B.1).

3) Remove the cable guard from the capstan.

4) Put capstan on test stand.

5) Place the adapter tool over the capstan and insert the positioning pin (from the straight-up position) to secure the adapter tool, as shown in Figure 2.

6) Insert a torque wrench (Snap On P/N TEP–6FUA or equivalent) as shown in Figure 2.

7) Apply 28 VDC (1 Amp max) power to the test stand jacks.

8) Using the capstan rotation switch on the test stand, turn the test stand motor on in the clockwise direction and read the torque on the wrench. Repeat this step in the counterclockwise direction.

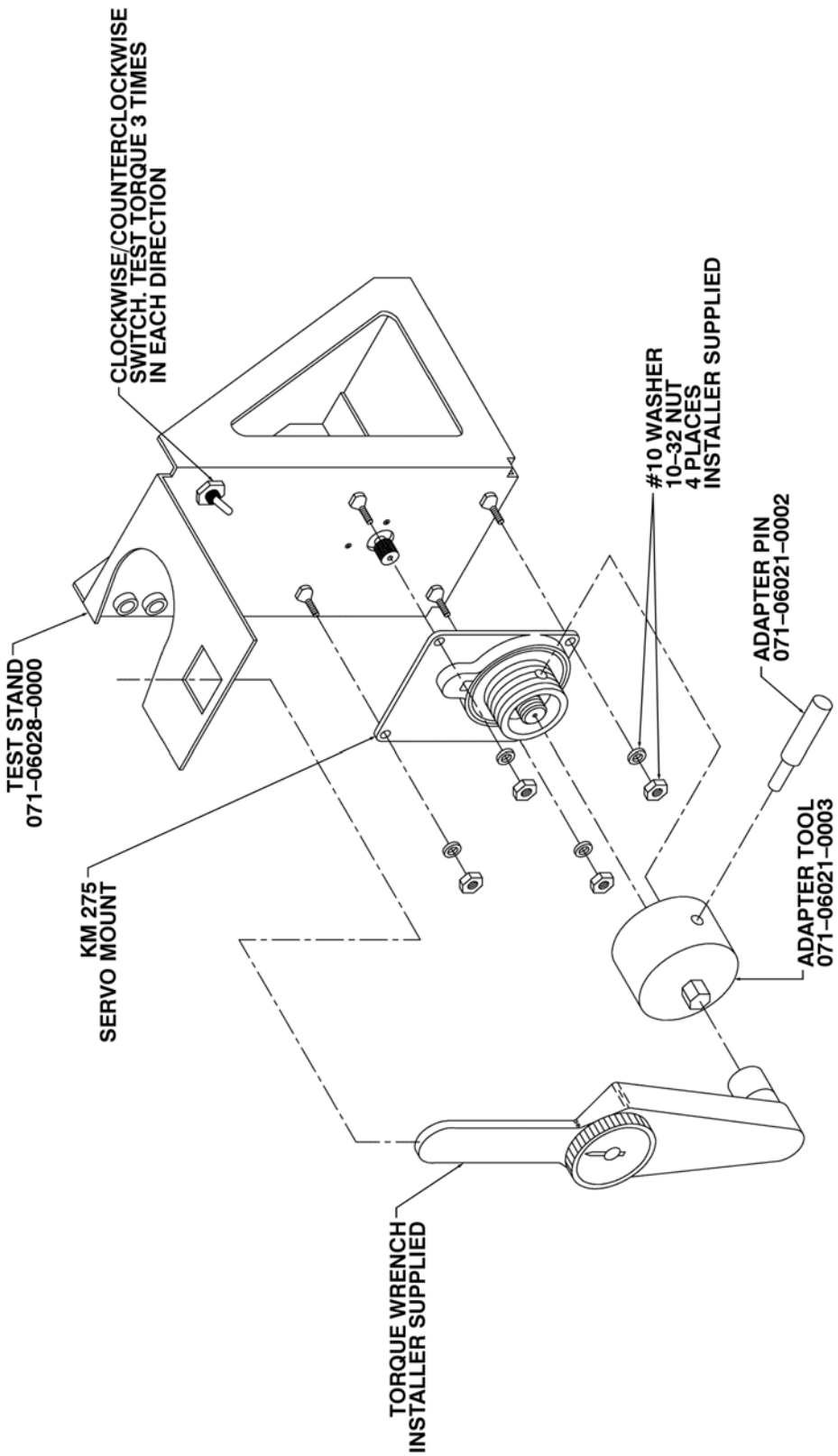
Note The desired reading is the average of the maximum and minimum readings obtained from the clockwise and counterclockwise rotations. The test is to be repeated 3 times in each direction and then average the 6 readings to determine the final torque reading. If the level determined falls below the desired value, rotate the clutch adjust nut clockwise. If the level determined falls above the desired value, rotate the clutch adjust nut counterclockwise. After any adjustment, repeat the torque test starting at Para 3.B.7).

Table 1 Trim Servo Clutch Setting

Item	Pitch Trim
Clutch Setting	22 in–lbs

9) Record the slip clutch torque reading, aircraft type, axis and date on the decal affixed to the servo mount body.

10) Reinstall the capstan, refer to Para 4.B.2).



KMC 275 Capstan Clutch Adjustment Bench Setup
Figure 2

Print Date: 2025-09-12

C. Bench Test KS 272A Pitch Trim Servo

CAUTION **ANY DISASSEMBLY/ASSEMBLY OF THE SERVO MUST BE DONE AT A STATIC-SAFE WORKSTATION. THE TECHNICIAN AND BENCH SHOULD BE GROUNDED.**

1) Remove the pitch trim servo from aircraft, refer to Para 4.A.1).

2) Remove the two screws that hold the cover of the unit, and carefully slide the cover off of the wiring harness.

3) Place the servo (without cover) on the bench so that the inner parts of the unit will not be damaged.

4) Manually move the solenoid with light pressure. The plunger should move freely in and out of the solenoid body.

   Note There should be no dirt, contamination or corrosion around the solenoid plunger. This could impede the actuation of the solenoid.

5) The release spring should pull the plunger out of the solenoid and against its stop freely and without hesitation.

6) Inspect the wiring for evidence of wear or damage to the insulation. This could cause wire shorts.

7) Inspect the entire servo for any loose hardware or other abnormalities.

8) Replace the cover by carefully sliding the cover back on over the wiring harness onto the unit in its original position.

9) Install the screws using low-strength thread-locking compound such as Loctite 222 or Loctite 242.

10) Reinstall the pitch trim servo, refer to Para 4.A.2).

D. Electrical Bonding

Refer to Figure 3 for details.

Conduct the following electrical bonding check:

Note If required, the maximum stated impedance values may be adjusted per the following: an acceptable impedance bond value (maximum milliohms) may be determined by multiplying the number of contact bond surfaces by 2.5 milliohms for each bond surface located between the probe contact points.

1) Servo Case to Airframe (Point 1 to Point 2)

   Remove non-conductive coating, if required, to achieve an acceptable contact point. Impedance bond value to be less than 2.5 milliohms. If not achievable, refer to Note.

2) Ground Module and Wire Shielding

   a) Ensure all DC power is disconnected (refer to 12-00-00, Servicing - General Data, Para 2.).

   b) Disconnect aircraft connector P278 from KS 272A Trim Servo wiring pigtail.

   c) If required, insert contact P/N M39029/31–228 or equivalent into P278 pin H for milliohmmeter attachment test point. Measure aircraft side of wiring harness between P278 pin H and airframe (point 2), impedance value to be less than 20 milliohms.

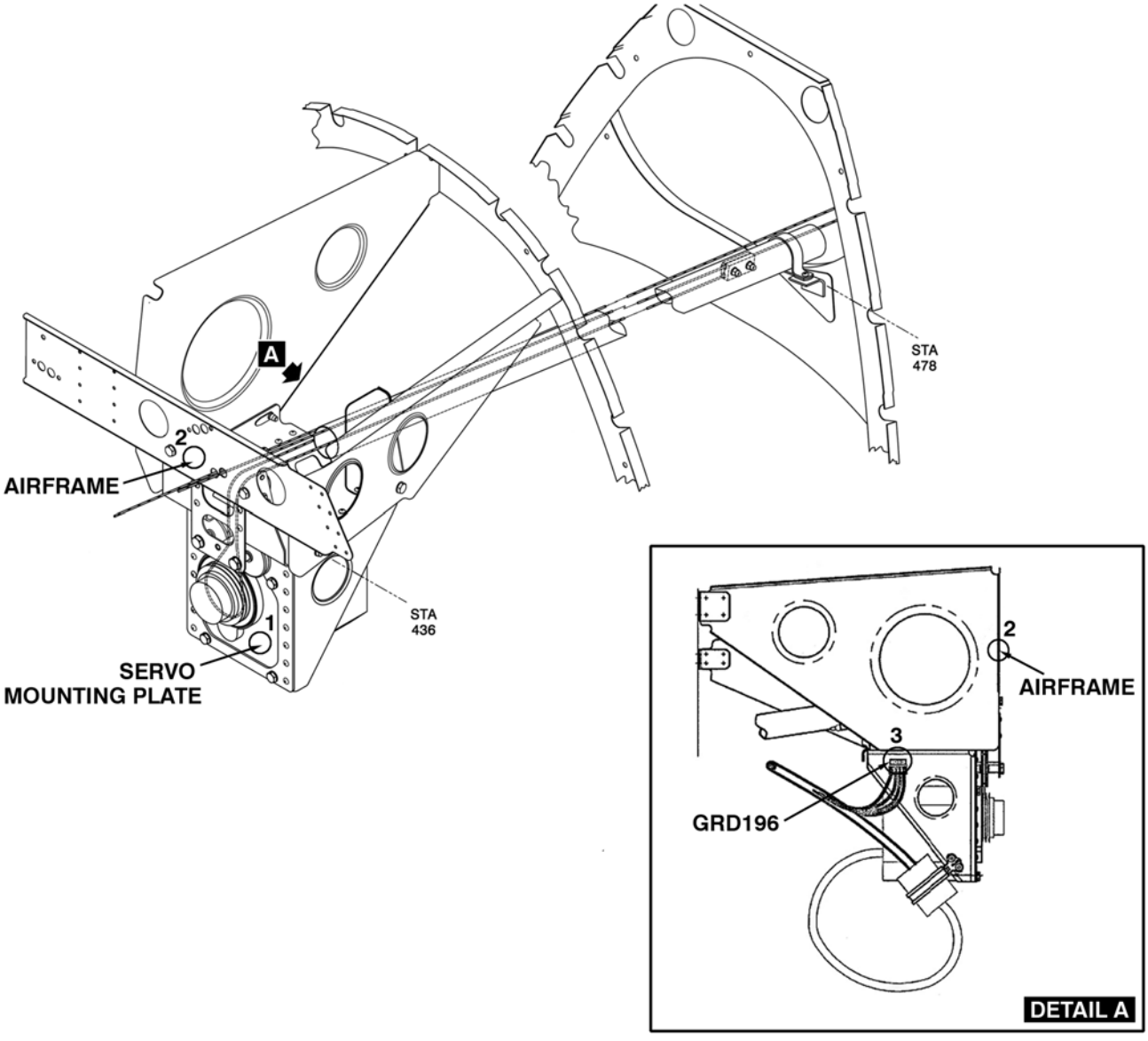
   d) If required, insert contact P/N M39029/31–228 or equivalent into P278 pin J for milliohmmeter attachment test point. Measure aircraft side of wiring harness between P278 pin J and airframe (point 2), impedance value to be less than 20 milliohms.

e) At wiring harness near P278, visually inspect all shield jumper wires between solder sleeves. Ensure all wires are secure in the solder sleeves.

f) At wiring harness near P278, visually inspect all shield drain wires between wiring harness P278 and GRD189. Ensure all shield drain wires are secure in the solder sleeves and GRD189.

g) Reconnect KS 272A Trim Servo connector P278 to KS 272A Trim Servo wiring pigtail.

h) Do Manual Electric Pitch Trim Operation Test, refer to Para 3.A.



Pitch Trim Servo Bonding Check
Figure 3

4. Removal/Installation

A. Pitch Trim Servo (KS 272A)

Refer to Figure 4 for details.

Note The KS 272A servo can be replaced without removing the capstan and disturbing the cable rigging.

1) Removal of the KS 272A Pitch Trim Servo

a) Ensure aircraft power is OFF. (Refer to 24-40-00, External Power – General Data).

b) Gain access to servo by removing bulkhead panel at rear of baggage compartment.

c) Disconnect servo electrical connector from receptacle.

d) Remove the two (2) bolts holding the pitch trim servo to the support bracket.

e) Holding the pitch trim servo securely, disengage the servo from the capstan.

f) Remove pitch trim servo from aircraft.

2) Installation of the KS 272A Pitch Trim Servo

WARNING INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED.

a) Ensure aircraft power is OFF. (Refer to 24-40-00, External Power – General Data).

b) Servo to airframe electrical bonding is maintained through direct contact between the servo and airframe mating surfaces. Prior to installation, ensure mating surfaces are clean and free of contaminants. As required, refer to procedures in 20-75-00, Electrical Bonding and Grounding of Aircraft Structures –General data.

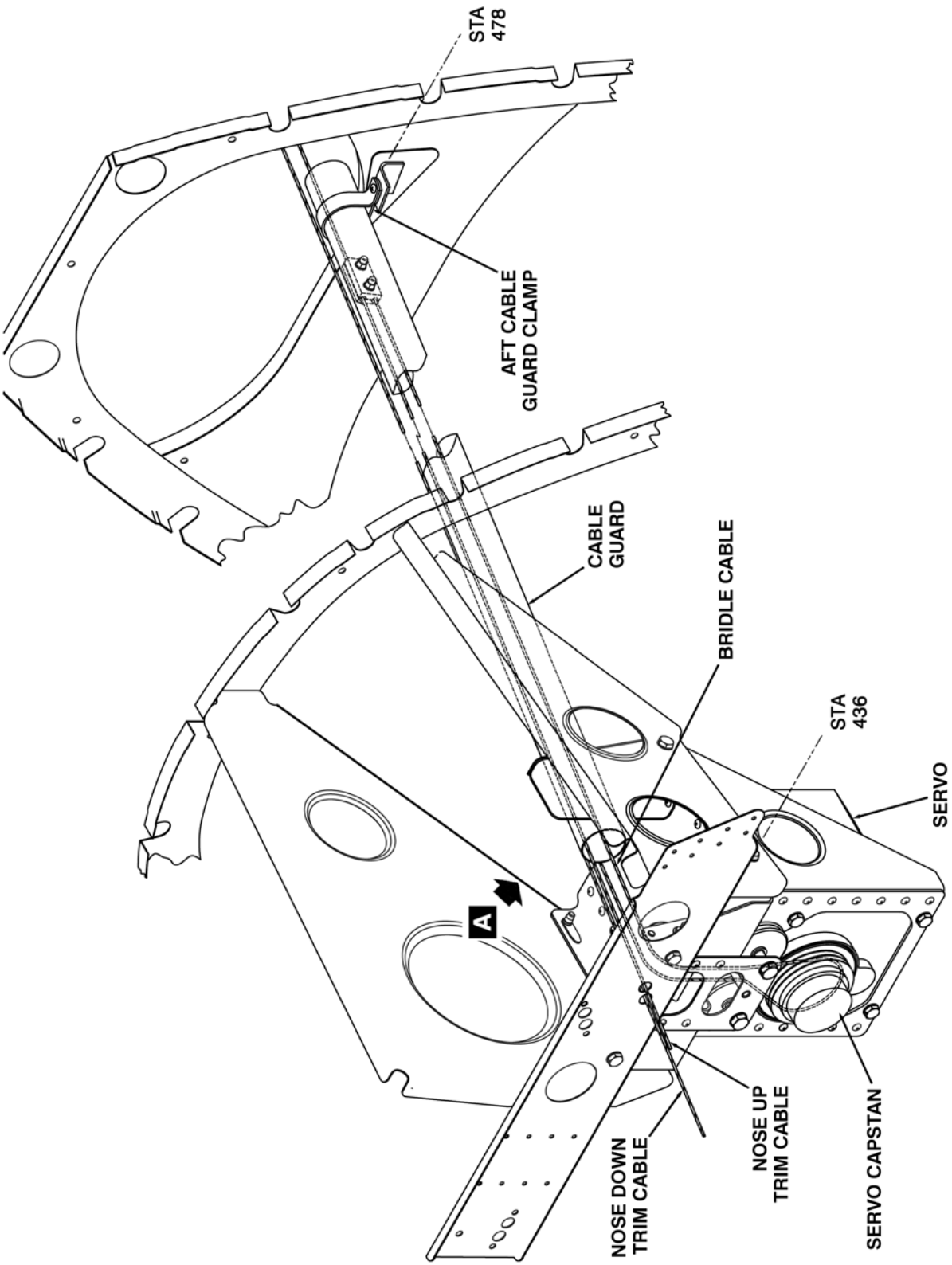
c) Engage servo drive shaft pinion with the teeth of the capstan gear.

d) Install the two (2) bolts to secure the servo to the support bracket.

e) Connect servo electrical connector to the receptacle, and ensure the locking clips are engaged.

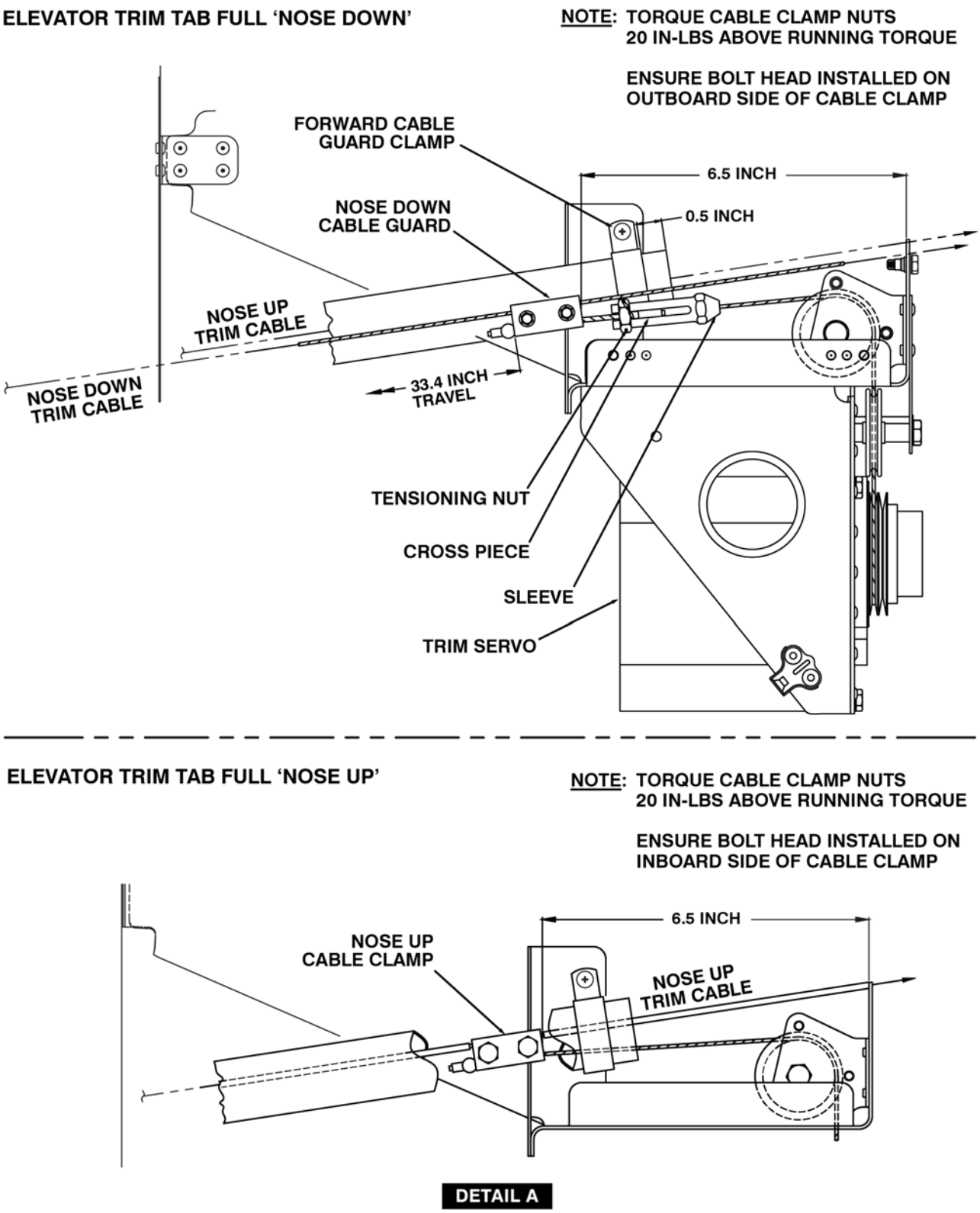
f) Do bonding check, refer to Para 3.D.

g) Function test system, refer to Para 3.A.



Pitch Trim Servo Installation
Figure 4 (Sheet 1 of 2)

Print Date: 2025-09-12



Pitch Trim Servo Installation
Figure 4 (Sheet 2 of 2)

B. Capstan (KM 275)

1) Removal of the KM 275 Capstan

a) Ensure aircraft power is OFF. (Refer to 24-40-00, External Power – General Data).

b) Remove the pitch trim servo, refer to Para 4.A.1).

c) Remove the cable guard from capstan pulley.

d) Remove locking wire from bridle cable tensioning nut and cross piece. Unscrew nut and disconnect cross piece from sleeve. Ensure enough slack in bridle cable to disengage cable from capstan.

e) Support capstan and remove the two (2) bolts securing the capstan to support bracket.

f) Remove capstan from aircraft.

2) Installation of the KM 275 Capstan

a) Ensure pitch trim capstan slip clutch has been calibrated in accordance with Para 3.B.

b) Ensure aircraft power is OFF. (Refer to 24-40-00, External Power – General Data).

c) Install the two (2) bolts securing the capstan to the support bracket.

d) Install servo bridle cable, refer to Para 4.C.2).

C. Pitch Trim Servo Bridle Cable

1) Removal of the Pitch Trim Servo Bridle Cable

a) Ensure aircraft power is OFF. (Refer to 24-40-00, External Power – General Data).

b) With elevator trim tab in full nose down position, remove locking wiring from bridle cable tensioning sleeve and cross piece. Unscrew tensioning nut and disconnect cross piece from sleeve.

c) Disconnect ‘nose down’ cable clamp.

d) With elevator trim tab in full nose up position, loosen forward and aft cable guard clamps and slide cable guard aft. Remove ‘nose up’ cable clamp.

e) Tension elevator trim cable. (Refer to 27-33-00, Elevator Trim Tab System – General Data).

f) Reposition cable guard and tighten securing clamps.

2) Installation of the Pitch Trim Servo Bridle Cable

a) Ensure aircraft power is OFF. (Refer to 24-40-00, External Power – General Data).

b) Loosen forward and aft cable guard clamps and slide cable guard aft.

c) Loop bridle cable around aft capstan pulley and clamp bridle cable to section of nose up trim cable with nylon sleeving removed. Torque clamp nuts to 20 in-lbs above running torque.

d) Reposition cable guard and tighten securing clamps.

e) Clamp bridle cable to nose down trim cable.

f) Ensure that the pitch trim control cables are correctly rigged. (Refer to 27-33-00, Elevator Trim Tab System – General Data).

g) Lock elevator controls in the neutral position.

h) Tension trim bridle cables in accordance with Table 2 — Trim Servo Bridle Cable Tension.

Note The aircraft must be allowed to stabilize at a temperature >55°F for a minimimum of 12 hours prior to adjustment of the bridle cable tension.

Table 2 Trim Servo Bridle Cable Tension

Item	Pitch Trim
Bridle Cable Tension	15 lbs + 2 lbs – 0 lbs

i) Lockwire tensioning nut and cross piece.

j) Install capstan cable guard.

k) Unlock elevator controls.

l) Manually cycle the electric pitch trim, ensure proper operation.

m) Function test system, refer to Para 3.A.

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